

DSC180 Q2 Project Proposal

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1 Context & Success Requirements

Modern digital platforms bury privacy and data-sharing settings behind complex interfaces, layers of web pages, and walls of text, making it difficult and frustrating for everyday users to understand what information they are agreeing to release and how to protect themselves.

Current tools mainly focus on static guidance or documentation for navigating these settings, not on automated, methodical change of privacy settings across multiple platforms.

Our team aims to create an integrated system that accomplishes exactly that combining:

- a chatbot capable of natural-language commands to facilitate changing settings;
- a centralized dashboard that visualizes settings across platforms; and
- agentic automation that executes changes on behalf of users.

A unified solution like this directly supports digital well-being and privacy literacy, providing a practical interface for users who lack time or technical understanding to manage settings across a variety of platforms. We have generated a structured database from privacy settings across platforms (Facebook, LinkedIn, Reddit, Zoom, Google, etc.) through our generalized scraping agent.

Planned tasks to support data availability include: schema design, integrated results, and visualization in a dashboard. We also plan to build out a secondary database based on successful test runs and user feedback, validating correctness and reproducibility. Additionally, use of a gold-standard manually verified baseline guarantees dataset integrity.

As far as computational feasibility, our dataset processing is accomplished through both human and agentic validation, and supported by our existing infrastructure (Chainlit, GitHub, structured JSON storage).

Functional success criteria:

- chatbot must reliably understand user commands and trigger correct system actions;
- dashboard must display useful information for comparison of settings across sites;
- database must support visualization, exploration, and batch-changing of settings; and
- chatbot and dashboard functionality should be merged into one coherent application.

2 Problem Statement & Previous Work

We must create a system that executes changes in accordance with user expectations, presents transparent visual representation, and is generalizable. Given a set of varying privacy-setting interfaces across multiple platforms, to build this system we need to construct:

- a structured dataset representing setting categories and states;
- a natural-language interface capable of mapping user instructions to setting modifi-

- cations; and
- a unified dashboard that visualizes these settings for comparison and access.

Previous work has established that the typical privacy notices given to users are often difficult to conveniently find and parse, meaning that they are often ignored [Schaub et al. \(2015\)](#). Additionally, notice designs are overly complex, lack real options for changing settings, and repetitive in terms of user agreements. There is progress on creating such a taxonomy as well as a design space for privacy notices, but at the moment it is not accessible to the ordinary users of services such as social media, for example. These users have a vested interest in knowing what information is being collected on them across services, but likely feel they lack time or capacity to pore over complex policies stored in different locations.

Through Agentic AI and simple visualizations, we aim to make a formerly academic and technical area of study tangible to the general public. At the moment, the best way for a user to understand the terms and conditions that they have agreed to is to comb through pages of text and settings, modifying them one by one if they are inclined to. In allowing users to access the privacy settings of many accounts a centralized location, we can mitigate the problem of disparate settings and accounts being inaccessible. The creation of such a service has broader benefits as well. Given a consistent taxonomy of privacy settings, designers, developers, and researchers could also be enabled to have a better understanding of the design preferences of their target audiences.

Aggregating trends and allowing users of all sorts of platforms to respond to what they learn has the auxiliary benefit of enabling privacy-centered design going forward. Designers' and software engineers' unawareness of what users expect and what they want can create numerous problems in the development life cycle [Li, Arya and Jin \(2024\)](#). Our investigation addresses deficiencies in existing solutions by providing a combination of visual exploration and automated action. This empowers users to gain greater understanding, while also simplifying workflows for ease of access. Key unique features of our work include an interactive dashboard linked to our privacy chatbot, as well as bulk privacy setting change, which allows users to modify their settings based on categories.

3 Primary Output Statement

The end goal is for users to have a one-stop-shop to comprehend their privacy settings, make decisions that suit their needs, and execute said decisions with ease and verification. The project will produce a fully functional web application consisting of:

- an AI-powered chatbot interface capable of natural-language commands to review and modify privacy settings;
- a centralized dashboard with interactive visualizations showing user settings for each platform, with comparison between platforms;
- a cross-platform privacy settings database; and
- integration of databases, dashboard with visualizations, and automated agent outputs into a single cohesive system.

As far as communicating our analysis, our dashboard will contain visualizations including:

- distribution of setting values
- recommended configurations
- platform-by-platform comparisons

The scraping agent produces data to be classified; this data will be analyzed by:

- categorizing by platform,
- labeling each setting under our data schema,
- supporting chatbot and dashboard pipelines.

References

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